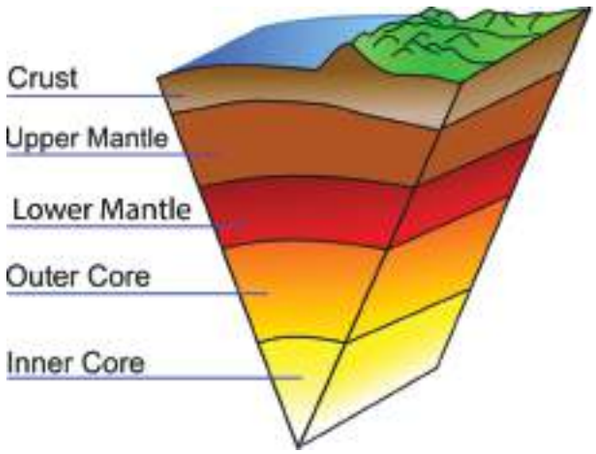
The lower part is a continuous zone of denser basaltic rocks forming the ocean floors, comprising mainly of silica and magnesium. It is therefore called Sima. It has an average density of 3.0g/cm³. The sial and the sima together form the earth's crust. Since the sial is lighter than the sima, the continents can be said to be 'floating' on a sea of denser sima.



Structure of the Earth

2. The Mantle

The next layer beneath the crust is called the mantle. It is separated from the crust by a boundary called Mohorovicic discontinuity. The mantle is about 2,900 km thick. It is divided into two parts: (i) Upper mantle, (ii) lower mantle. The density of the mantle is 3.7g/cm³.

Why the interior of the earth is so hot?

3. The Core

The innermost layer of the earth is called the core. It is also known as *barysphere*. It is separated from the mantle by a boundary called *Weichart-Gutenberg discontinuity*. The core is about 2900km - 6370km thick. The core is also divided into two parts.

(i) **The outer core,** which is rich in iron, is in liquid state.

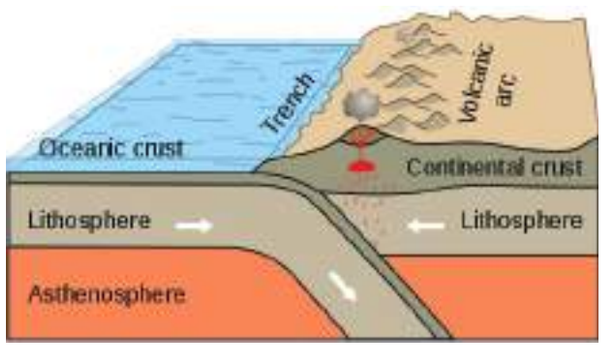
(ii) **The inner core**, composed of Nickel and Ferrous *(Nife)*, is solid in state. The central core has very high temperature and pressure. The average density of core is 13.0 g/cm³.

DO YOU KNOW?

The crust forms only 1% of the volume of the earth, 84% consists of the mantle and 15% makes the core. The radius of the earth is 6,370km.

The Earth Movements

The lithosphere is broken into a number of plates known as the *Lithospheric plates*. Each plate, oceanic or continental moves independently over the Asthenosphere. The movement of the Earth's lithospheric plates is termed as tectonic movements. The energy required to move these plates is produced by the internal heat of the earth. These plates move in different directions at different speed.



Lithospheric Plates

At places, these plates move away from each other creating wide rifts on the earth's surface. At some places, these plates come closer and collide. When an oceanic plate collides with a continental plate, the denser oceanic plate is forced below the continental plate. As a result of the pressure from above the rocks heats up and melts. The molten rocks rise again forming volcanic mountains along

7th_SS_Geography_Unit_1.indd 146 23-01-2025 12.14.40 PM